

Regular Expressions (Regex) in Python

1 Introduction

Regular Expressions (Regex) ek powerful technique hai jo text processing me use hoti hai. Iska main purpose text ke andar specific patterns ko search karna, extract karna, replace karna aur manipulate karna hota hai. Python me Regex ka use `re` module ke through hota hai.

Roman Urdu me samjhen to: Regex ka matlab hai "text ke andar koi pattern dhoondhna". Jaise ke numbers, emails, words ya koi specific format.

2 Basic Regex Concepts

- `\d` = digits (0-9)
- `\w` = word characters (letters + digits + underscore)
- `\s` = spaces
- `.` = koi bhi character
- `+` = ek ya zyada baar repeat

Roman Urdu Explanation: Yeh symbols regex ke building blocks hain. Inko combine karke hum patterns banate hain.

3 `re.findall()` – Sab Matches Extract Karna

Yeh function text me jitne bhi matches hote hain un sab ko list me return karta hai.

Examples

Example 1

```
import re
re.findall(r"\d+", "Mera number 123 hai")
```

Explanation: Yeh code text me jitne bhi numbers hain unko nikalta hai. `\d+` ka matlab hai ek ya zyada digits.

Example 2

```
re.findall(r"\d+", "12 aur 45 dono hain")
```

Explanation: Yeh multiple numbers ko alag alag extract karta hai.

Example 3

```
re.findall(r"\w+", "Hello Python 123")
```

Explanation: Yeh words aur numbers dono ko ek sath detect karta hai.

Example 4

```
re.findall(r"[a-zA-Z]+", "Hello123")
```

Explanation: Sirf alphabets ko match karta hai, numbers ignore ho jate hain.

Example 5

```
re.findall(r"\s", "Hi there")
```

Explanation: Yeh space character ko detect karta hai.

Example 6

```
re.findall(r"\w+@\w+\.\w+", "Email: abc@gmail.com")
```

Explanation: Yeh ek simple email pattern detect karta hai.

Example 7

```
re.findall(r"[A-Z]+", "HELLO world")
```

Explanation: Sirf capital letters match hote hain.

Example 8

```
re.findall(r"[a-z]+", "HELLO world")
```

Explanation: Sirf small letters match hote hain.

Example 9

```
re.findall(r"\W+", "Hello@123")
```

Explanation: Yeh special characters detect karta hai jaise @.

Example 10

```
re.findall(r"\w+", "abc_123")
```

Explanation: Underscore bhi word character hota hai is liye include hota hai.

4 re.sub() – Replace Karna

Yeh function kisi pattern ko kisi aur value se replace karta hai.

Examples

Example 1

```
re.sub("cat", "dog", "cat is cute")
```

Explanation: "cat" ko "dog" se replace kiya gaya.

Example 2

```
re.sub(r"\d+", "", "abc123")
```

Explanation: Numbers remove kar diye gaye.

Example 3

```
re.sub(r"\s+", "", "a b c")
```

Explanation: Spaces remove kar diye.

Example 4

```
re.sub(r"\d", "*", "123")
```

Explanation: Har digit ko * se replace kiya.

Example 5

```
re.sub(r"\S+@\S+", "", "mail abc@gmail.com")
```

Explanation: Email address remove kar diya.

Example 6

```
re.sub(r"[a-z]", "X", "abc")
```

Explanation: Small letters ko X se replace kiya.

Example 7

```
re.sub(r"[A-Z]", "x", "ABC")
```

Explanation: Capital letters ko x se replace kiya.

Example 8

```
re.sub(r"\W", "", "hi@#")
```

Explanation: Special characters remove kiye.

Example 9

```
re.sub(r"\s+", " ", "a b")
```

Explanation: Multiple spaces ko single space me convert kiya.

Example 10

```
re.sub(r"\d+", "", "abc123xyz")
```

Explanation: Beech ke numbers remove kiye.

5 re.split() – Text Split Karna

Yeh function text ko pattern ke basis pe tod deta hai.

Examples

Example 1

```
re.split(",", "a,b,c")
```

Explanation: Comma pe split karta hai.

Example 2

```
re.split("[,;]", "a,b;c")
```

Explanation: Multiple separators use karta hai.

Example 3

```
re.split(r"\s+", "a b c")
```

Explanation: Spaces ke basis pe split.

Example 4

```
re.split(r"\d+", "abc123xyz")
```

Explanation: Numbers ke basis pe split.

Example 5

```
re.split("-", "2023-01-01")
```

Explanation: Date ko split kiya.

Example 6

```
re.split(r"\W+", "hello@world!")
```

Explanation: Special characters pe split.

Example 7

```
re.split(":", "time:12:30")
```

Explanation: Colon pe split.

Example 8

```
re.split("a", "banana")
```

Explanation: Har 'a' pe split.

Example 9

```
re.split("x", "text")
```

Explanation: 'x' pe split.

Example 10

```
re.split(r"\.", "a.b.c")
```

Explanation: Dot pe split.

6 Conclusion

Regex ek powerful tool hai jo text cleaning, pattern matching aur data extraction ke liye use hota hai. Yeh NLP aur data preprocessing me bohat important role play karta hai.